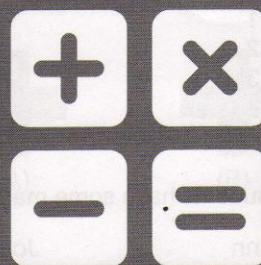




UNSW Global
THE UNIVERSITY OF NEW SOUTH WALES
SYDNEY • AUSTRALIA

**PAPER
A**



2012
ICAS

International Competitions
and Assessments for Schools

MATHEMATICS

**Educational
Assessment
Australia**
eaa.unsw.edu.au

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

40 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your
NAME, SCHOOL and **OTHER INFORMATION**.
Use a 2B or B pencil.
Do **NOT** use a pen.
Rub out any mistakes completely.

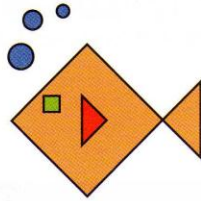
You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.
Your score will be the number of correct answers.
Marks are **NOT** deducted for incorrect answers.

There are **40 MULTIPLE-CHOICE QUESTIONS** (1–40).
Use the information provided to choose the **BEST** answer from
the four possible options.
On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a ruler and spare paper.
You are **NOT** allowed to use a calculator.

1. Mike drew this picture.



How many triangles are there in his picture?

- (A) 1
- (B) 2
- (C) 3
- (D) 7

2. Four students have some marbles.

Ann



Joe



Lin



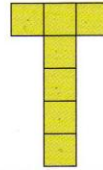
Mike



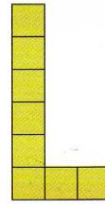
Which student has a different number of marbles from the others?

- (A) Ann
- (B) Joe
- (C) Lin
- (D) Mike

3. Which shape has the smallest area?



(A)



(B)



(C)



(D)

4. Part of this photo is missing.



Which option shows the missing part of the photo?



(A)



(B)



(C)



(D)

5. At a school athletics carnival, students win a medal for coming first or second in a race.
- Mike came first in three races and second in two races.
- How many medals did Mike win?
- (A) 2
(B) 3
(C) 4
(D) 5

6. What is the value of 2 in the number 2491?
- (A) 2
(B) 20
(C) 200
(D) 2000

7.

$$42 \quad ? \quad 6 = 7$$

Which operation sign is missing?

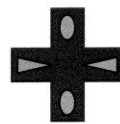
- (A) +
(B) -
(C) ×
(D) ÷

8. Joe planted 3 rows of trees, with 8 trees in each row.

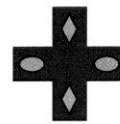
How many trees did Joe plant altogether?

- (A) 24
(B) 11
(C) 8
(D) 3

9. Which design has a shape on it that is not on any of the others?



(A)



(B)



(C)



(D)

10. Fadi asked the students in his school year to choose their favourite food from a list. Each student chose only one favourite food.

The tally table shows the results.

Favourite food	Tally
steak	
pasta	
pizza	
salad	
sushi	

How many students did Fadi ask?

- (A) 18
(B) 42
(C) 45
(D) 50

11. Lin placed blocks on grid squares as shown.

	A	B	C	D	E	F	G
1						■	
2		■	■	■		■	
3						■	
4						■	
5				■		■	
6	■	■		■			
7				■			

Which of these squares has a block on it?

- (A) F6
(B) D5
(C) C3
(D) A1

12. This is part of a calendar from the year 2007.

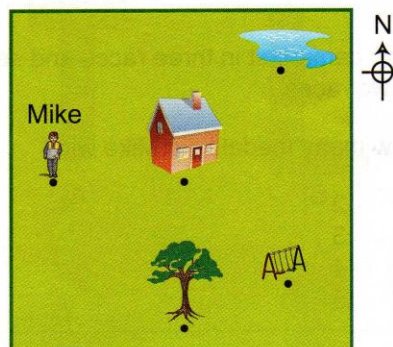
MAY 2007						
SUN	MON	TUE	WED	THU	FRI	SAT
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

Fadi's birthday is on the 15th of May.
Lin's birthday is 10 days later.

On what day of the week was Lin's birthday in 2007?

- (A) Tuesday
(B) Thursday
(C) Friday
(D) Sunday

13. From where Mike is standing, he walks in a straight line to the tree.



In which direction does Mike walk?

- (A) north-east
(B) north-west
(C) south-east
(D) south-west

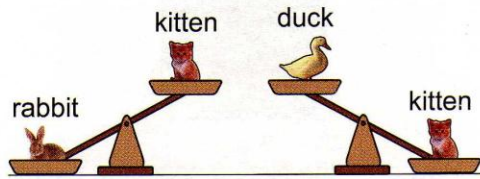
14. Sasi replaced the $\triangle$ in this calculation with the number 10.

$$6 \times \triangle - 2 = \square$$

What number is $\square$?

- (A) 12
(B) 48
(C) 58
(D) 60

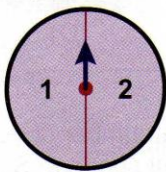
15. Mike is comparing the mass of his three pets.



Which option lists Mike's pets in order from biggest mass to smallest mass?

- (A) duck, kitten, rabbit
- (B) kitten, rabbit, duck
- (C) rabbit, duck, kitten
- (D) rabbit, kitten, duck

16. Which spinner has the greatest chance of landing on 2?



(A)



(B)

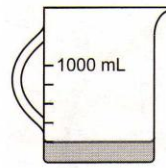


(C)

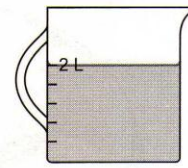


(D)

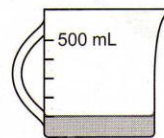
17. Which jug contains exactly 200 mL of liquid?



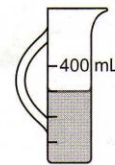
(A)



(B)

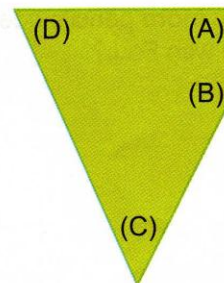


(C)

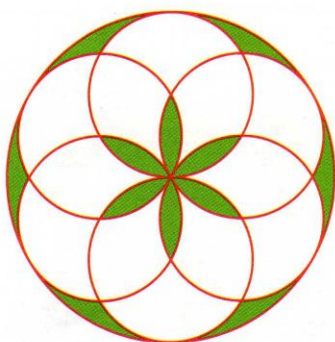


(D)

18. Which angle inside this shape is a right angle?



19. How many complete circles have been drawn to make this pattern?



- (A) 6
(B) 7
(C) 12
(D) 13

20. Fadi has 21 pencils. Joe has 46 pencils.

How many **more** pencils does Joe have compared with Fadi?

- (A) 25
(B) 26
(C) 46
(D) 67

21. Bella holds this picture in front of a mirror.



How does the picture appear in the mirror?



(A)



(B)

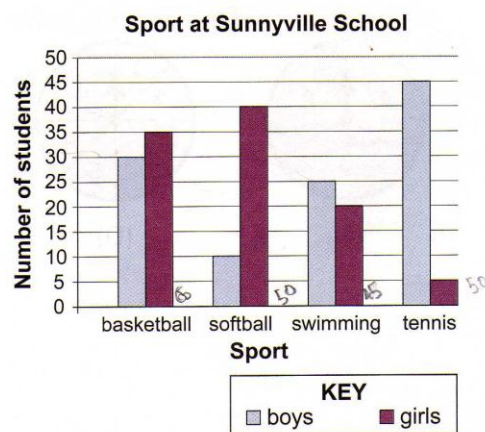


(C)



(D)

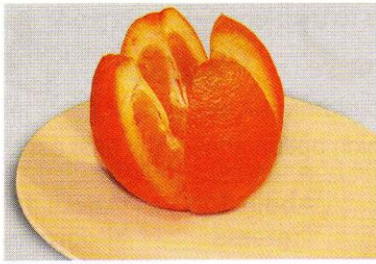
22. This graph shows the number of boys and girls who take different sports at Sunnyville School.



Which sport is taken by the least number of students?

- (A) basketball
(B) softball
(C) swimming
(D) tennis

23. Manling, Sunil and Adam ate some oranges that had been cut into quarters.



Manling ate five orange quarters.
Sunil ate six orange quarters.
Adam ate nine orange quarters.

How many whole oranges were eaten altogether?

- (A) 4
(B) 5
(C) 10
(D) 20

24. Joe built a model using a blue block, a red block and a yellow block.

This is what his model looks like from the front.



What does Joe's model look like from above?



(A)



(B)

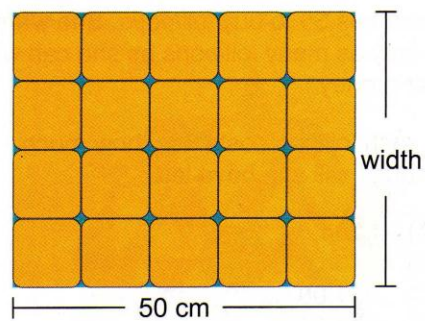


(C)



(D)

- 25.



What is the width of the rectangle?

- (A) 4 cm
(B) 5 cm
(C) 40 cm
(D) 50 cm

26. This is a number pattern.

209
228
247
266
285

?

323
342
361
380

What is the missing number?

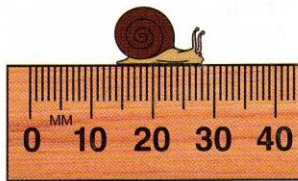
- (A) 294
(B) 296
(C) 304
(D) 306

27. Noor has \$5 to buy lollipops. She wants to buy as many lollipops as she can with her money.

If each lollipop costs 80c, how much change will she have left?

- (A) 20c
- (B) 40c
- (C) \$1.00
- (D) \$4.20

28. A snail was crawling along a ruler.



Fully stretched, the end of the snail's tail was on the 14 mm mark and its head just reached the 28 mm mark on the ruler.

Then the snail crawled to the right and stretched itself again. The end of its tail was now on the 18 mm mark.

What mark on the ruler did the snail's head reach?

- (A) 24 mm
- (B) 32 mm
- (C) 34 mm
- (D) 36 mm

29. Lin and her friends estimated the mass of a banana, a car and a baby.



Which option shows the best estimates?

	About 100 grams	About 10 kilograms
(A)		
(B)		
(C)		
(D)		

30. Sunil had a 20 m roll of masking tape. He cut off 15 cm.

How much tape is left on the roll, in cm?

- (A) 1985
- (B) 500
- (C) 185
- (D) 5

36. Ann is building towers using just blue and yellow blocks.

She can build only four different towers that are two blocks high.



How many different towers can she build that are three blocks high?

- (A) 6
- (B) 7
- (C) 8
- (D) 9

37. Joe was given this subtraction.

$$\begin{array}{r} 2 \text{ ? } 6 \\ - 1 \text{ 3 } 7 \\ \hline 1 \text{ 1 } 9 \end{array}$$

What is the missing number?

- (A) 2
- (B) 3
- (C) 4
- (D) 5

38. Lin and Ann made paperclip chains that were the same length.

Lin used four large paperclips and two small paperclips.



Ann used two large paperclips and five small paperclips.

$2L + 5S$



Mike made a chain that was the same length as Lin's chain and Ann's chain joined together.

Mike used only small paperclips.

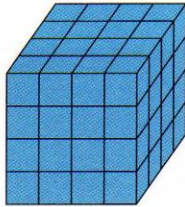
How many small paperclips did Mike use?

- (A) 7
- (B) 8
- (C) 13
- (D) 16

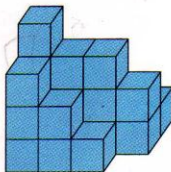
39. Fadi is using some small cubes



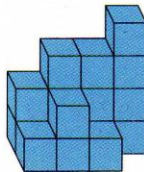
to build a large cube like the one below.



The pictures show the view from the front and the view from the back of what he has built so far.



view from
the front

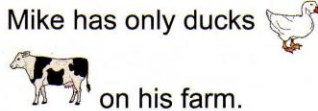


view from
the back

How many more small cubes does Fadi need to complete the large cube?

- (A) 26
- (B) 37
- (C) 42
- (D) 48

40. Mike has only ducks and cows



on his farm.

The cows and ducks have 52 legs in total.
There are 4 more cows than ducks on
Mike's farm.

How many ducks are there on Mike's
farm?

- (A) 26
- (B) 13
- (C) 10
- (D) 6

**M****2012 Mathematics Answer Keys****ICSA**International Council
and Assessments for

Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Papers G & H	Paper I
1	B	A	D	D	D	C	A	
2	B	A	C	A	B	A	C	
3	A	D	B	C	A	C	C	
4	A	D	B	B	B	A	B	
5	D	B	A	A	D	B	B	
6	D	D	D	C	D	B	A	
7	D	C	A	B	A	B	A	
8	A	B	B	D	B	A	D	
9	B	D	A	C	C	A	C	
10	D	B	A	D	D	D	B	
11	B	B	C	A	B	A	C	
12	C	B	B	A	A	B	B	
13	C	A	A	D	B	D	D	
14	C	A	B	B	A	D	C	
15	D	C	D	D	C	C	B	
16	A	A	C	D	B	C	C	
17	A	C	C	B	D	B	C	
18	A	C	D	A	C	A	A	
19	B	B	D	C	A	D	A	
20	A	B	A	B	D	C	C	
21	D	D	C	B	B	A	D	
22	C	A	A	D	C	C	C	
23	B	C	D	B	C	C	B	
24	D	B	B	B	B	B	B	
25	C	C	B	A	A	D	B	
26	C	A	D	C	A	C	D	
27	A	C	C	A	C	C	B	
28	B	C	A	B	B	A	B	
29	B	D	C	A	B	C	D	
30	A	B	C	C	B	C	A	

(Please



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M | 2012 Mathematics Answer Keys

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Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Papers G & H	Papers I & J
31	C	C	C	A	C	B	D	D
32	C	D	D	A	C	D	B	D
33	A	C	A	C	C	B	D	D
34	A	B	A	B	A	D	A	C
35	B	C	D	D	A	D	D	A
36	C	D	D	72 072	5 05 005 0 5	16 016	4 04 004 0 4	26 026
37	D	A	B	10 010	600	566	3 03 003 0 3	9 09 009 0
38	D	A	C	90 090	32 032	27 027	4 04 004 0 4	39 039
39	C	C	C	5 05 005 0 5	9 09 009 0 9	137	9 09 009 0 9	75 075
40	D	B	B	500	900	6 06 006 0 6	127	126

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ICAS 2012 MATHS ANSWERS – PAPER A

1. B
2. B
3. A
4. A
5. D
6. D
7. D
8. A
9. B
10. D
11. B
12. C
13. C
14. C
15. D
16. A
17. A
18. A
19. B
20. A
21. D
22. C
23. B
24. D
25. C
26. C
27. A
28. B
29. B
30. A
31. C
32. C
33. A
34. A
35. B
36. C
37. D
38. D
39. C
40. D